



Habib University

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Course Title: **LINEAR ALGEBRA**

Instructors:

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Class ID: **MATH 205**

Total Marks: **100**

Duration: **3 hours**

	Marks Obtained	Instructions for Students: a. Read the following instructions carefully. b. Please ensure that your mobile phone is switched off. c. All your personal belongings must be placed at the designated area of the examination hall. d. Write your Name, ID, Subject, Section, Instructor's name, and date in the space provided below. e. Remember to sign the exam attendance sheet circulated by the head invigilator / proctor. f. Answer all questions in the exam. g. You must provide complete working and reasoning for all steps of your solution. h. If you need to leave the room for any reason, please raise your hand and take permission from head invigilator / proctor. i. After finishing your exam, raise your hand and give your answer book to the head invigilator / proctor.
Q. 1		
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Instructor's Signature & Date		

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MATH 205 LINEAR ALGEBRA
FINAL EXAM SPRING 2023



Question 1

- (a) Prove the following theorem: If A is any matrix, then the row space and column space of A have the same dimension. [7]

- (b) For a square matrix A , show that A and A^T have the same rank. [3]



Question 2

- (a) Show that a system of linear equations $A\underline{x} = \underline{b}$ is consistent if and only if $\underline{b}$ is in the column space of A . [7]

- (b) Using (a), explain why, if T_A is the linear transformation given by pre-multiplication of a column vector $\underline{x}$ by the matrix A , then the dimension of the range of T_A is equal to the rank of A . [Hint: See if you can understand why the range is equal to the column space of A] [3]



Question 3 Let V be an inner product space. Let $\mathbf{u}, \mathbf{v} \in V$. Prove that: (*Note: These are for any general inner product spaces. Do not assume that the space or inner product is Euclidean*)

(a) $|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \|\mathbf{v}\|$ [5]

(b) $\langle \mathbf{u}, \mathbf{v} \rangle = \frac{1}{4} \|\mathbf{u} + \mathbf{v}\|^2 - \frac{1}{4} \|\mathbf{u} - \mathbf{v}\|^2$ [5]



Question 4 Let $S = \{v_1, v_2, \dots, v_r\}$ be a set of vectors in the Euclidean space $\mathbb{R}^n$. Show that if $r > n$, then S is not linearly independent. **[10]**



Question 5 (a) Use the inner product $\langle \mathbf{f} | \mathbf{g} \rangle = \int_0^1 f(x)g(x)dx$ to compute $\langle \mathbf{f} | \mathbf{g} \rangle$, for the vectors $\mathbf{f} = \cos 2\pi x$ and $\mathbf{g} = \sin 2\pi x$. [4]

(b) Let V be the vector space spanned by $\mathbf{v}_1 = \cos^2 x$, $\mathbf{v}_2 = \sin^2 x$, $\mathbf{v}_3 = \cos 2x$.

(i) Show that $\mathcal{S} = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is not a basis for V . [3]

(ii) Find a basis for V , giving clear reasoning for your answer. [3]



Question 6 Indicate whether each statement is always True, or sometimes False. Justify your answer by giving a logical argument or a counterexample: **[2 each]**

(a) If A is a diagonalizable matrix, then the rank of A is the number of nonzero eigenvalues of A .

(b) If $\underline{v}$ is any $n \times 1$ matrix, and I is the $n \times n$ identity matrix, then $I - vv^t$ is orthogonally diagonalizable.

(c) Let $T: M_{nn} \rightarrow R$ be the transformation that maps an $n \times n$ matrix into its determinant, i.e. $T(A) = \det(A)$. Then T is a linear transformation.

(d) If $T: V \rightarrow W$ is a linear transformation, then T is one-to-one iff $\ker(T) = \{\mathbf{0}\}$.

(e) Every possible linear transformation between Euclidean spaces can be represented by some matrix.



Question 7(a) Prove: If k is a positive integer, λ is an eigenvalue of matrix A , and X is a corresponding eigenvector, then prove that λ^k is an eigenvalue of A^k and X is a corresponding eigenvector. [5]

(b) Hence or otherwise, if $A = \begin{bmatrix} 4 & 0 & 1 \\ -2 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$, find the eigenvalues of A^{100} . [5]



Question 8 Let $S = \{v_1, v_2, \dots, v_n\}$ is an orthogonal set of nonzero vectors in an inner product space, then set S is linearly independent. **[10]**



Question 9 For the given matrix $A = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & -3 & -2 & 0 \\ 10 & -5 & 15 & 10 & 0 \\ 22 & -10 & 18 & 8 & 0 \end{bmatrix}$, find a basis for the row space of A consisting entirely of row vectors from A . **[10]**



Question 10 (a) Consider the rotation matrix in $\mathbb{R}^2$. For a rotation of angle θ , the trace is $2 \cos(\theta)$. Moreover, the determinant of this matrix must be 1. However, if $0 < \theta < \pi$ then prove that this matrix has no eigenvalues. [5]

(b) Can you give a geometric explanation for (a)? [2]

(c) For matrices A , B , and C , prove that if A is similar to B , and B is similar to C , then A is similar to C . [3]



ROUGH WORK



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